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Jacobian Scopes: A Unified Geometric Framework for Token-Level LLM Attributions

 (/pdf?id=x0pAg7jich)

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TL;DR: We propose Jacobian Scopes, a suite of gradient-based token attribution methods for LLMs unified by a single geometric framework: each scope is a vector-Jacobian projection onto a geometrically motivated direction in hidden space.

Abstract:

Gradient-based attribution methods for large language models (LLMs) have so far been limited to explaining scalar outputs - the logit of a single target token. Yet LLM predictions are inherently distributional, and many practically important questions concern the full predictive distribution: which tokens make the model uncertain? Which drive a broad range of plausible continuations? We propose **Jacobian Scopes**, a unified framework that fills this gap by projecting the input-to-output Jacobian onto different directions in output space via a single vector-Jacobian product, requiring only one backward pass. This yields three complementary methods: **Semantic Scope** attributes a specific target logit; **Fisher Scope**, grounded in information geometry, identifies tokens that most alter the overall shape of the predicted distribution; and **Temperature Scope** traces which tokens govern the model's predictive confidence. Fisher and Temperature Scopes are, to our knowledge, the first attribution methods to natively target distributional rather than pointwise features of LLM predictions. Through case studies spanning instruction following, translation, and in-context time-series forecasting, Jacobian Scopes reveal implicit political biases, uncover word- and phrase-level translation strategies, and illuminate the nearest-neighbor pattern-matching mechanisms underlying LLM forecasting. Quantitative evaluation on LAMBADA and IWSLT2017 across six leading LLMs (LLaMA-3.2, Qwen2.5, Gemma-3) confirms that Jacobian Scopes consistently match or outperform Input × Gradient and Integrated Gradients, at a fraction of the latter's computational cost.

Checklist Confirmation:  I confirm that I have included a paper checklist in the paper PDF.

Responsible Reviewing:  We acknowledge the responsible reviewing obligations as authors.

Primary Area: Language and multimodal language models (e.g., text generation, summarization, VQA)

Secondary Area: Socio-technical aspects of AI (e.g., fairness, interpretability, privacy, safety, governance)

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Strengths And Weaknesses:**Strengths:**

1. Casting token attribution as a single VJP $v_{T,t}$ and obtaining the influence score as the ℓ_2 norm of one auto-differentiated scalar loss $L = v_T y$ is elegant and correct. It cleanly subsumes plain gradient saliency as the $v = w_{\text{target}}$ case and makes the "choice of v = choice of explanandum" idea explicit and extensible.
2. Fisher Scope and Temperature Scope attribute features of the whole predictive distribution and of predictive confidence rather than a single logit. To my knowledge, this distributional framing of token attribution is new and is the paper's main contribution.
3. A single VJP with parameter gradients disabled gives 0.027s backward vs 0.069s forward on an A4000, i.e. attribution at roughly the cost of a forward pass, versus the multi-step path integration of Integrated Gradients.
4. The observation that Fisher/Temperature Scopes, which never consult the ground-truth token, match or exceed target-aware methods under AOPC

Weaknesses:

1. AOPC gains are mostly within or near the reported error and are not significance-tested. Many "match or outperform" claims rest on small differences against a standard error of (+-) 0.01. No paired significance test across the passages is reported, so the claim of "consistently match or outperform" is not statistically substantiated for most cells. The main claim that distributional methods beat target-specific ones is therefore weakened.
2. Faithfulness is assessed only by AOPC with zero-vector ablation. There is no comprehensiveness/sufficiency decomposition and, more importantly, no saliency sanity check.
3. AOPC ablates tokens by replacing embeddings with zero vectors, yet the paper attributes Integrated Gradients' weaker scores partly to distortions/attention-sink effects of the all-zero baseline. The same OOD concern plausibly biases the AOPC metric used to rank all methods; this should be addressed.
4. Treating discrete vocabulary token values v as continuous (to derive) is a strong assumption for general LLM vocabularies; it is reasonable for the numeric-token ICL setting, but the scope of validity should be stated explicitly, as Temperature Scope's interpretation as "inverse variance" depends on it.

Quality: 3: good**Clarity:** 3: good**Significance:** 3: good**Originality:** 3: good**Questions:**

- Does Proposition 1's continuous-token approximation hold (or degrade gracefully) for non-numeric vocabularies, or is Temperature Scope's variance interpretation intended to be claimed only for the structured numeric-ICL regime?
- Can paired significance tests (bootstrap or Wilcoxon) be reported for the AOPC differences in Table 1? In how many of the 12 model x dataset cells is the best Jacobian Scope significantly better than the strongest non-Scope baseline?
- Given that paper attributes IG's weakness partly to the all-zero baseline, does replacing zero-vector ablation with a mask-token or marginalisation baseline preserve the ordering among methods?

Limitations:

Yes.

Some more discussion should be present on: (i) the statistical fragility of the AOPC comparisons, and (ii) the evidential status of the qualitative interpretive claims, especially the political-bias example.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

N/A

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Official Review of Submission30235 by Reviewer UMqk

Official Review by Reviewer UMqk 📅 25 Jun 2026, 21:21 (modified: 23 Jul 2026, 12:46)

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📄 Revisions (/revisions?id=3wAnDrrbzE)

Summary:

This paper introduces Jacobian scopes, a unified attribution framework for LLMs that focuses on both individual token predictions and the model's entire predictive distribution. The framework projects the input-to-output Jacobian onto different output-space directions, yielding three attribution methods: semantic for target token, Fisher for output distribution, and temperature for model confidence. The framework is evaluated using AOPC on LAMBADA and IWSLT2017.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

The paper introduces a unified framework for token attribution based on vector-Jacobian products and proposes two new methods, Fisher and Temperature scopes. The general idea is very interesting. However, the main weaknesses lie in the evaluation of the method and in the paper's presentation.

First, the paper does not sufficiently motivate the importance of this problem. In the introduction, it is assumed that token-level attribution improves understanding of LLM behavior. However, attribution quantifies local input sensitivity and does not necessarily explain the computations or reasoning processes of the model.

Additionally, the empirical evaluation does not support its claims. The qualitative examples are cherry-picked, and while they help understand how the method works, they do not really help evaluate it. The quantitative analysis uses AOPC to determine whether removing high-ranking input tokens reduces the probability of the ground-truth token. While this is a standard faithfulness metric for saliency methods, it does not show that the proposed methods provide a better understanding of model behavior or reasoning, nor does it evaluate whether the generated outputs meaningfully change after attribution-guided interventions.

Finally, the interpretation of the quantitative results is not convincing. The paper argues that target-specific methods should have an inherent advantage and interprets the comparable/higher performance of Fisher and Temperature Scope as evidence that distribution-level attribution is superior. However, the model's predictions depend only on the input and model parameters, not on the ground-truth token used during evaluation. Consequently, it is unclear why target-aware methods should be expected to outperform, making this conclusion insufficiently supported.

As a side note, the paper's organization makes it difficult to read. Several theoretical justifications and empirical analyses appear in the appendix, while the main paper contains unnecessarily large figures for the method and qualitative examples. As a result, reading the paper requires frequent cross-referencing between the main text and the appendix.

Quality: 2: not good

Clarity: 2: not good

Significance: 2: not good

Originality: 3: good

Questions:

Can you elaborate more on why a gradient-based attribution is needed in NLP? I believe this motivation should be added to the introduction. The introduction should convince the reader why this is important and why they should continue reading the paper.

Please include additional evaluation methods to study the model's actual output, or alternatively, revise the claims and framing to better align with the presented methodology and experiments.

The paper inconsistently uses both "Figure" and "fig." to reference figures. It also uses both "section X" and "Appendix X" when referring to the appendix. Please choose one style for each and consistently use it throughout the paper.

Limitations:

yes

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

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NA

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Official Review of Submission30235 by Reviewer D5fu



Official Review by Reviewer D5fu 12 Jun 2026, 05:24 (modified: 23 Jul 2026, 12:46)

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Summary:

The paper addresses LLM interpretability, more precisely by presenting a suite of gradient-based causal attribution methods called Jacobian Scopes. Several explanation/attribution objectives can be covered with the approach: Semantic Scope asks which input tokens drive the predicted probability of a specific target word; Fisher Scope asks which tokens most change the overall shape of the predicted distribution; and Temperature Scope asks which tokens control the model's confidence, i.e., how sharply peaked the distribution is. The authors evaluate the relevance of their method across several case studies in natural language instruction, as well as in non-NLP settings such as time-series forecasting. They then evaluate attribution faithfulness quantitatively on two benchmark datasets, LAMBADA and IWSLT2017, across 3 LLMs: LLaMA-3.2, Qwen2.5, and Gemma-3.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

-The paper goes beyond standard gradient-based attribution methods by exploiting the full predictive distribution of LLMs, rather than attributing only a single scalar output such as one token logit. -Although I'm not an expert in interpretability methods, paper is clear and relatively easy to follow -According to experiments of section 5, distribution-level attribution can be as faithful than target-specific gradient attribution, while remaining useful in settings where there is no single obvious target token.

Weaknesses:

-Limited scope of impact: while the paper introduces a useful extension of gradient-based attribution to distributional properties of LLM outputs, the contribution remains within the relatively narrow setting of local input-output sensitivity analysis. -While Jacobian Scopes, especially Fisher Scope, consistently improve over Input × Gradient in Table 1, the absolute gains are often small and the paper does not report paired significance tests.

Quality: 3: good

Clarity: 3: good

Significance: 3: good

Originality: 3: good

Questions:

implicit questions provided in my review

Limitations:

yes

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.

Confidence: 2: You are willing to defend your assessment, but it is quite likely that you did not understand the central parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

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